ASSIGNMENT 2

TITLE: PERIPHERAL GEOGRAPHIES, GLOBAL SYSTEMS: ARTIFICIAL INTELLIGENCE AS SOCIO-TECHNICAL IMPERATIVE FOR ISLAND COMMUNITIES

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ABSTRACT

Artificial intelligence is frequently framed as a purely technical breakthrough, yet its development, deployment, and impacts are deeply embedded in social relations, institutional structures, financial mechanisms, and contested power dynamics. This essay examines AI through a socio-technical systems lens, arguing that neither pure technological advancement nor social organization alone can adequately explain AI's trajectory or its differential impacts across geographies. Drawing on lived experiences situated in the Andaman and Nicobar Islands, a geographically remote context with unique demographic composition and colonial-historical legacies, this analysis interrogates how the AI revolution proceeds as though universal high-speed internet, reliable electricity, and access to elite institutions are given, while peripheral communities experience stark infrastructure deficits. The essay addresses conceptual foundations of socio-technical systems and AI; philosophical and historical contexts revealing accumulated geographic inequalities; the concentration of massive financial investments and firm-oriented model customization in wealthy centers; contested narratives surrounding the "AI bubble"; the mechanisms through which algorithmic governance replicates existing power asymmetries; and the fundamental requirement for geographic equity and infrastructural development in achieving meaningful AI governance. Through complementarity perspective and grounded first-person observations reflecting internet access challenges, student life experiences, and government's rapid technological integration without addressing systemic gaps, this work reveals how AI's real-world performance and societal impact depend fundamentally on alignment between technical systems and social-geographic contexts.

Keywords: Artificial Intelligence, Socio-Technical Systems, Geographic Marginalization, Technological Infrastructure, Algorithmic Governance, Peripheral Development, Complementarity, Island Geographies.

1. INTRODUCTION

I am writing this essay from my subjective understanding, lived experiences and observations around how AI models have penetrated aspects of society previously human. Last week, I sat in my room in Port Blair (Sri Vijaya Puram), attempting to download a dataset for my AI research on environmental vulnerability assessment. The internet connection, unreliable as always on these islands, kept dropping, forcing me to restart the download multiple times. The connection speeds barely exceeded 2 Mbps, while I knew my experiences back in Mumbai, working with the same AI systems having access to fiber optic networks delivering gigabits per second. This seemingly technical inconvenience reflects something far more fundamental: **The architecture of technological infrastructure that enables or constrains who can meaningfully participate in the AI revolution**.

Contemporary discourse around artificial intelligence oscillates between utopian promises of superhuman intelligence and dystopian fears of mass unemployment or human extinction. These polarized narratives often treat AI as if it were an autonomous entity developing according to internal technical logic. What they obscure is how AI systems are produced and governed through complex interactions among engineers, venture capitalists, technology firms, regulators, workers, and end-users, and crucially, how this system of production privileges certain geographies, certain communities, and certain ways of knowing while marginalizing others. The central problem this essay addresses is therefore: **How can AI be understood as a Socio-Technical system, and what does this perspective reveal about its development, uses, and possible futures, particularly when viewed from peripheral, geographically isolated communities?**

From a complementarity perspective, AI should be understood neither as a looming replacement of human capacities nor as a mere tool subordinate to human will, but as **a set of systems that co-evolve with social institutions, worker practices, regulatory frameworks, infrastructure systems, and cultural meanings**. For those of us in the islands, this co-evolution reveals stark inequalities: the AI revolution proceeds as though universal high-speed internet, reliable electricity, and access to elite institutions are givens. Yet when your student life involves rationing data downloads, when ferries fail to connect you with the teachers and collaborators you need, when your research depends on computational resources controlled by distant corporations, the universality of this "revolution" becomes questionable.

2. CONCEPTUAL FOUNDATIONS

2.1. Defining Socio-Technical Systems

Socio-technical systems theory emerged in the mid-twentieth century to analyze how technical and social elements function together as integrated wholes rather than as separable layers. A socio-technical system comprises a **configuration** of human actors, technologies, organizational routines, institutional rules, and cultural meanings that function together to produce outcomes. The theory emphasizes that technologies are not neutral instruments; they embody specific interpretations of problems and preferred solutions, yet the same technical artifact can support vastly different social arrangements depending on how it is embedded into organizations, regulatory regimes, and geographic contexts.

Early research, particularly studies of coal mining operations, discovered that technological systems' performance depended critically on social organization and worker autonomy. Researchers found that implementing new machinery without redesigning work organization and worker relationships led to productivity failures, a phenomenon termed "complementarity failure". When organizations achieved success, it invariably involved "**joint optimization**", designing both technical systems and social arrangements to work smoothly together.

Three primary subsystems characterize socio-technical configurations: (i) the **social subsystem** comprising people, relationships, values, hierarchies, and organizational structures; (ii) the **technical subsystem** comprising tools, algorithms, hardware, standards, and control functions; and (iii) the **environmental subsystem** encompassing the broader regulatory, cultural, and infrastructural context, including geographical, climatic, and demographic realities, within which both operate. These subsystems are characterized by mutual causality: social factors influence technology use and adaptation, while technical affordances enable, constrain, and shape what social arrangements are feasible.

2.2. Defining AI: Beyond Technical Specifications

Artificial intelligence is broadly defined as the field of computer science concerned with building systems that exhibit behaviours traditionally regarded as requiring intelligence if performed by humans, including perception, reasoning, learning, language use, and problem-solving. Yet as Geoffrey Hinton, recognized as a pioneering architect of deep learning, emphasizes, AI represents not simply mathematical procedures but systems capable of

detecting patterns, learning from experience, and performing tasks previously requiring human cognition.

However, this technical achievement exists inextricably embedded within organizational, financial, and social structures. OpenAI did not emerge from mathematical insight alone but from **specific venture capital configurations**, recruitment of specialized talent, institutional partnerships, and regulatory environments permitting large-scale computational experimentation. Anthropic similarly represents not a pure technical innovation but a reorganization of AI talent motivated by safety concerns, a fundamentally socio-technical reconfiguration. For those of us outside these centers of capital and talent accumulation, this raises immediate questions: **Where do peripheral geographies fit in this globally organized system of AI production?**

From my experience as in the islands attempting to engage with contemporary AI, I observe that AI is simultaneously and inseparably: (i) Mathematical techniques (neural networks, transformer architectures, backpropagation algorithms); (ii) Specialized hardware systems (GPUs, TPUs, neuromorphic chips) requiring massive infrastructure investment that remains inaccessible on remote islands; (iii) Training data reflecting societal patterns, biases, and power structures predominantly drawn from data-rich, affluent regions; (iv) Organizational arrangements distributing labour between humans and machines, but distributing this labour unequally across geographies; (v) Financial mechanisms allocating billions toward specific technical directions, predominantly in wealthy nations; and (vi) Regulatory, ethical, and institutional frameworks constraining and enabling development, often designed without input from peripheral communities.

2.3. The Complementarity Principle Applied to AI: A Geographic Perspective

Complementarity describes relationships between institutions, practices, or systems wherein "**one institution functions all the better that some other particular institutions or forms of organisation are present**". When complementary elements align, they reinforce each other and generate coherence; when misaligned, the entire system exhibits dysfunction despite any individual component's technical sophistication.

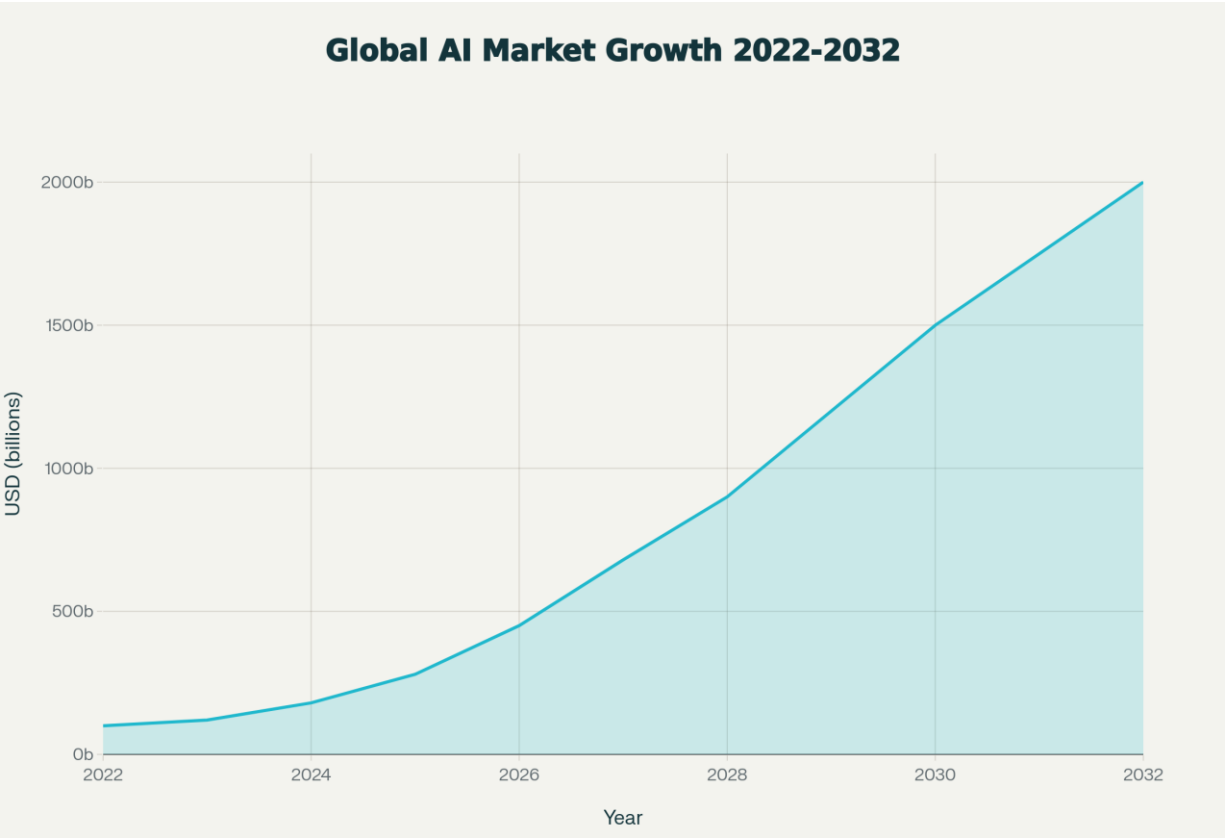
Consider a case that resonates with my own reality: A machine learning model for disaster prediction, designed to forecast monsoon-driven flooding or cyclone impacts, exhibits exceptional technical accuracy in laboratory conditions trained on global climate data. Yet it

fails to improve outcomes in the Andaman and Nicobar Islands. It cannot integrate with the actual disaster response infrastructure, which relies on ferry-dependent communication often disrupted during the very weather events being predicted. Local officials and tribal communities distrust algorithmic recommendations from models trained on data from entirely different geographical and climatic contexts. Regulatory frameworks designed for mainland scenarios prohibit full implementation of algorithmic recommendations. Incentive structures reward predictions that minimize costs to government infrastructure rather than prioritizing protection of peripheral communities.

Here, technical excellence proves insufficient; success requires complementary social and organizational arrangements that are deeply attentive to geography, local governance, and community participation.

From my observation of how the government attempts to rapidly integrate AI-driven technologies on the islands without addressing systemic gaps, complementarity failures are pervasive. The Administration has launched initiatives like digital administration portals, algorithmic systems for welfare distribution, and technology-based service delivery, yet these systems assume reliable internet access (which remains intermittent for most island residents); they require digital literacy and documentation (challenging for tribal and elderly populations); they deploy algorithms designed for mainland administrative contexts (inadequate for island-specific governance challenges); and they implement these systems without meaningfully consulting the communities they ostensibly serve or addressing the underlying infrastructure deficits.

This is precisely the kind of complementarity failure visible when technological integration proceeds faster than social and infrastructural adaptation.



Global AI Market Growth Trajectory (2022-2032): Projected Market Size

Generated using Perplexity AI

3. PHILOSOPHICAL AND HISTORICAL CONTEXTS

3.1. The Question of Whose Intelligence Gets Valued

The philosophical foundations of AI trace to Alan Turing's revolutionary 1950 paper "Computing Machinery and Intelligence," wherein Turing posed the deceptively simple question: "Can machines think?" Rather than becoming entangled in philosophical definitions of "thinking," Turing proposed a pragmatic test: if a machine could produce responses indistinguishable from a human's, attributing "thinking" to the machine would be reasonable.

However, Turing's genuine contribution extended beyond the famous test. His paper articulated three distinct pathways toward machine intelligence: (i) AI through explicit programming of rules; (ii) AI through machine learning from experience; and (iii) AI through combining logic, probabilities, learning, and background knowledge. Turing recognized that the first approach would prove limited machines could not be programmed with rules for every contingency. Instead, he proposed what he termed "the child machine"—a machine that learns from experience the way human infants do.

From my perspective, Turing's genius lay not merely in technical insight but in his profound recognition that machine intelligence would require complementarity between multiple approaches combining formal logic, probabilistic reasoning, learning mechanisms, and domain knowledge. This insight remains vital for contemporary AI development, and it raises a crucial question often unexamined: whose domain knowledge gets incorporated into machine learning systems?

In the context of the Andaman and Nicobar Islands, this question becomes acute. The tribal communities, Shompen, Nicobarese, Onge, Andamanese and Jarawa, possess profound domain knowledge about island ecosystems, sustainable resource use, climate adaptation, and disaster response refined over millennia. Yet when AI systems are deployed for environmental management, resource allocation, or disaster prediction, they are trained on data that almost never includes indigenous knowledge systems. Instead, these systems incorporate scientific paradigms, remote sensing data, and models developed in temperate and tropical mainland contexts. The implicit message is that machine intelligence should incorporate every form of human knowledge except that held by peripheral, traditionally marginalized communities.

Contemporary philosophy grapples with persistent tensions: the problem of consciousness (whether machines could possess subjective experience), the gap between narrow AI

(addressing specific tasks) and general AI (matching human cognitive flexibility), and the relationship between intelligence and understanding. These philosophical questions prove significant not as abstract puzzles but because they inform how societies should approach AI governance, ethics, and regulation, and from whose perspectives and for whose benefit those governance decisions are made.

3.2. From Symbolic AI to Deep Learning- A Story of Accumulated Inequalities

The history of AI reveals cyclical patterns of enthusiasm and disillusionment, each enabled and constrained by socio-technical contexts. Yet this history has been written almost exclusively from the perspective of wealthy nations and elite institutions. The Dartmouth Conference of 1956 represented a pivotal socio-technical moment, bringing together researchers from disparate fields (Computer Science, Mathematics, Physics) with unified commitment to exploring "synthetic intelligence". This conference institutionalized AI as a legitimate academic discipline, established research labs at MIT, Carnegie Mellon, and Stanford, and inaugurated the Turing test as a central benchmark for progress. What is less often acknowledged: this conference was held in a wealthy nation with abundant computing resources, electrical infrastructure, and research funding. The countries and regions lacking such resources were simply absent from this founding moment.

Early optimism proved naive. The "**AI winter**" of the 1980s resulted not from philosophical confusion but from a concrete complementarity failure: available computational power, data quantity, and algorithmic sophistication proved inadequate for transforming theoretical possibilities into practical systems. Social enthusiasm for AI initially outpaced technical capabilities and hardware availability, creating a classic pattern wherein hype generates investment, early disappointments trigger skepticism, and investment subsequently contracts.

For nations and regions like India and the Islands within it, this history meant that crucial developmental periods passed before local AI capabilities could be built. By the time deep learning breakthroughs emerged in the 2010s, the technological infrastructure was already concentrated in Silicon Valley, Beijing, and other wealthy centers. Starting from an infrastructure deficit has compounding consequences.

The emergence of **Machine Learning** in the **1980s** and **1990s** represented a socio-technical reorientation. Rather than attempting to explicitly encode human knowledge, researchers increasingly pursued statistical learning from data. This shift aligned with 3 crucial

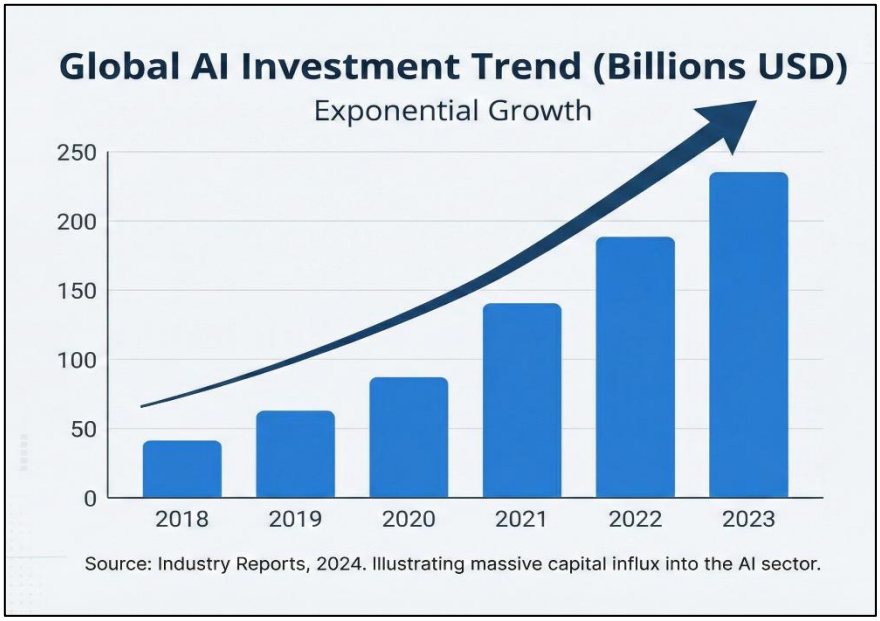
developments: accumulation of digital data as computing systems proliferated, growth of computational capacity enabling statistical analysis at scale, and theoretical breakthroughs in backpropagation enabling deep neural networks. Yet who was generating this digital data? Predominantly people in wealthy countries with digital infrastructure, digital literacy, and digital services. The data that would train the AI systems of the future was being created unequally across geographies.

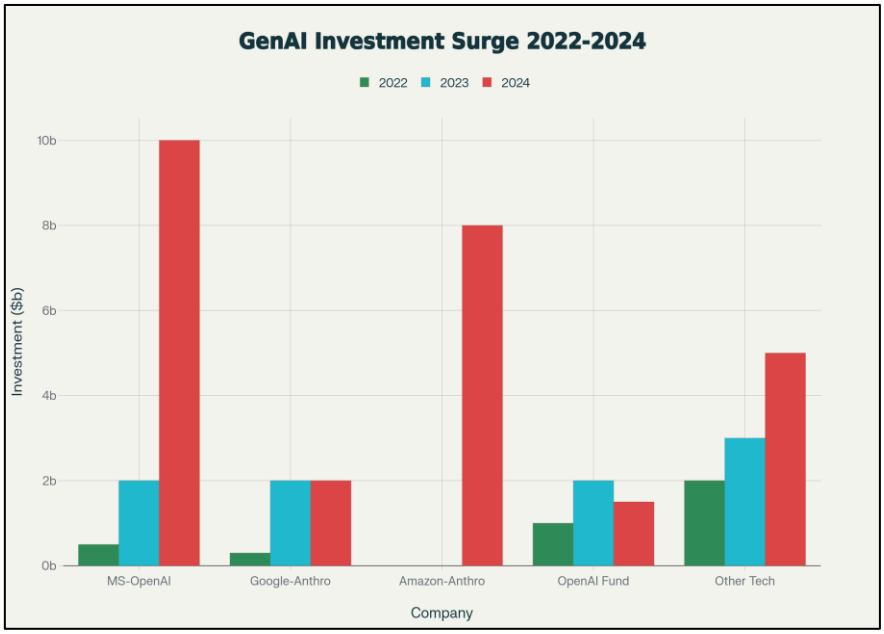
Deep learning's contemporary dominance emerged from a specific socio-technical conjuncture. Geoffrey Hinton's breakthroughs with neural networks proved technologically significant, but their real-world transformation required: massive computational infrastructure (GPUs optimized for parallel processing), availability of large labelled datasets (reflecting organizational digitization, concentrated in wealthy nations), specialized talent trained in mathematics and computer science (concentrated in elite universities and technology firms), and institutional frameworks permitting long-term research investment without immediate commercial returns. Most crucially, all these prerequisites were already in place in wealthy countries. In peripheral regions, they remain only partially developed.

4. THE FINANCIAL ARCHITECTURE OF AI

4.1. Massive Investment and Corporate Competition

The contemporary period has witnessed AI financing of unprecedented scale. Market-research estimates suggest that the global AI market size stood at roughly USD 100 billion in 2022, rising to approximately USD 180-250 billion by 2024, with projections surpassing USD 1.5-2 trillion by 2032. Private investment in generative AI specifically has increased dramatically, from under USD 1 billion in 2022 to tens of billions by 2024.





Global AI Investment Trend and Generative AI Investment Surge

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At the firm level, companies such as **OpenAI**, **Anthropic**, and **Google** have attracted multi-billion-dollar investments and strategic partnerships. Microsoft committed USD 10 billion to OpenAI in 2023. Google invested up to USD 2 billion in Anthropic, while Amazon committed up to USD 8 billion to Anthropic by late 2024.

From analyzing these investment patterns, I observe they represent a profound institutional concentration: large technology platforms recognized that alternative AI paradigms could fragment their market positions. By investing in competitors like Anthropic, these platforms simultaneously: (i) Hedge against risks that superior AI systems might emerge outside their control; (ii) Establish partnership arrangements generating recurring infrastructure revenue; (iii) Capture equity gains as valuations appreciate; and (iv) Influence competitive trajectories through resource allocation. Notably absent from this ecosystem are investment flows toward peripheral regions, indigenous communities, or developing nations seeking technological autonomy.

The semiconductor industry reflects this concentration acutely. The global AI semiconductor market was valued at USD 60.6 billion in 2024 and is projected to reach USD 169.4 billion by 2032. This explosive growth reflects not merely demand for AI computing but fundamental restructuring of hardware requirements. Traditional CPUs optimized for sequential processing prove inefficient for neural network operations.

Yet the islands have virtually zero local semiconductor manufacturing capacity, and likely never will at current global economics. This means AI development on the islands will remain entirely dependent on importing expensive hardware and software controlled by distant corporations. This is a form of technological colonialism, an updated version of historical patterns wherein peripheral regions provide raw materials and markets but control over technology remains centralized.

4.2. Firm-Oriented Model Customization

Beyond massive aggregate investments, the AI sector simultaneously experiences intensive firm-specific model customization. Unlike previous technology cycles where firms purchased standardized software and adapted it to needs, contemporary organizations increasingly develop custom machine learning models aligned to their specific objectives, data, and business processes.

Custom machine learning development represents a fundamentally socio-technical endeavour. Organizations implementing custom models achieve approximately 60% improvement in predictive accuracy compared to standard solutions. However, this improvement does not arise solely from superior algorithms but from profound organizational redesign: custom models require deep problem definition in collaboration with stakeholders, careful data collection reflecting organizational contexts, feature engineering incorporating domain knowledge, and iterative refinement with continued business input.

Large technology firms have resources for model customization specific to their corporate needs. Firms on the mainland can hire data scientists, collect proprietary data, fine-tune models for their specific operational contexts. But what about the Andaman and Nicobar Islands administration, tribal welfare organizations, or fisheries cooperatives? They have no resources for customization. They must use whatever off-the-shelf AI systems are available, which means importing systems optimized for mainland or global contexts, not for island-specific governance challenges.

From my conversation with an administrator working on welfare distribution systems in the islands, I learned that the government attempted to deploy an algorithmic allocation system developed for mainland welfare distribution. The system made assumptions about documentation availability (tribal populations often lack formal identity documents), communication infrastructure (assumes reliable phone/internet access), and demographic patterns (uses national averages rather than island-specific distributions). The system was technically sophisticated but contextually catastrophic. Yet there were no resources to customize it, no local data scientists to adapt it, no framework for iterative improvement with community input.

This exemplifies how firm-oriented customization creates institutional lock-in effects that disadvantage peripheral regions: once an organization invests in building custom models trained on proprietary data, it becomes dependent on internal AI capabilities and specialists. Large firms with resources for substantial R&D and specialist hiring increasingly pull away from smaller competitors and peripheral regions unable to justify customization investments. The result is a system wherein technological capability becomes yet another axis of regional inequality.

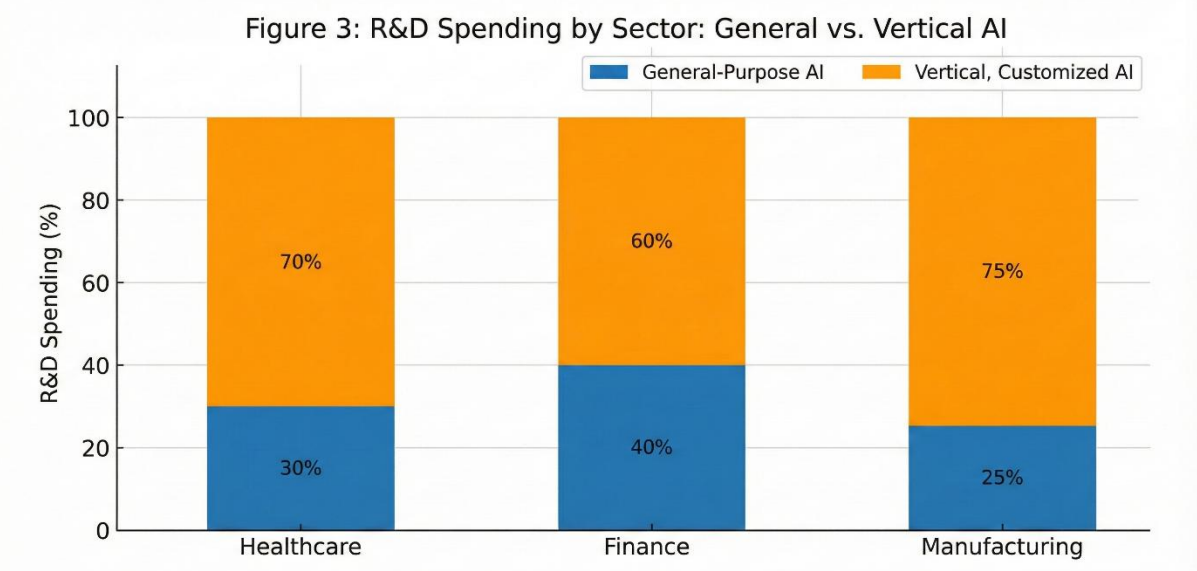

Figure 3: R&D Spending by Sector: General vs. Vertical AI. This chart shows the proportion of R&D spending allocated to general-purpose AI versus vertical, customized AI solutions in key industries.

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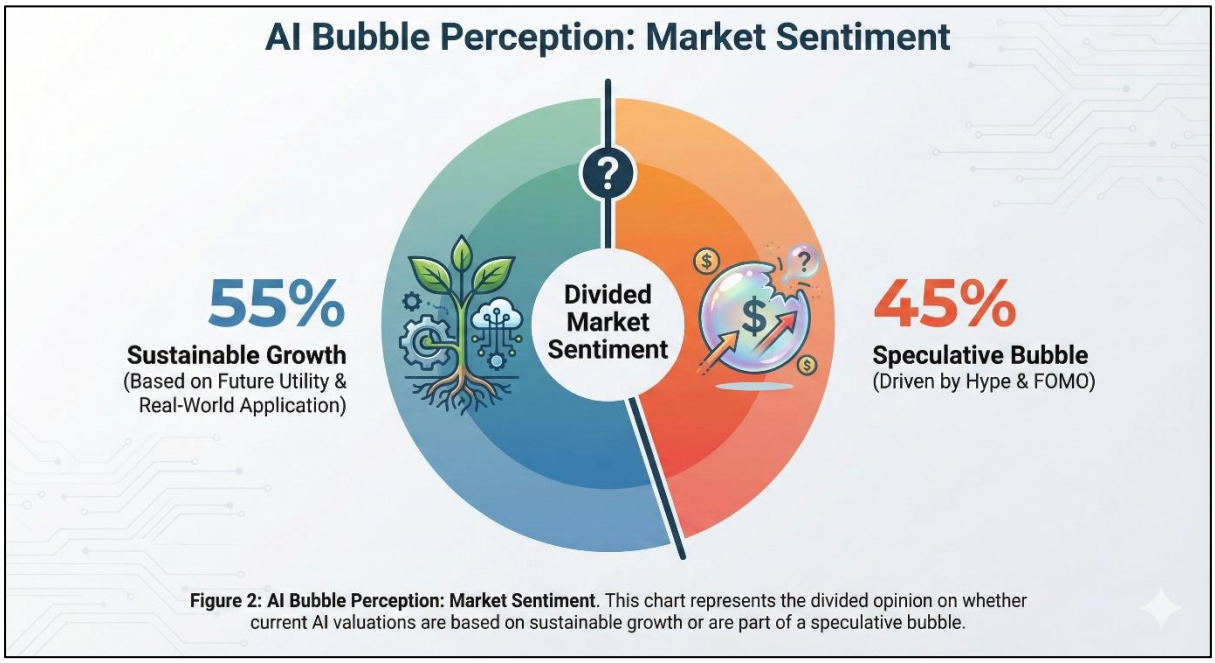
4.3.The "AI Bubble" Narrative: A Perspective from the Periphery

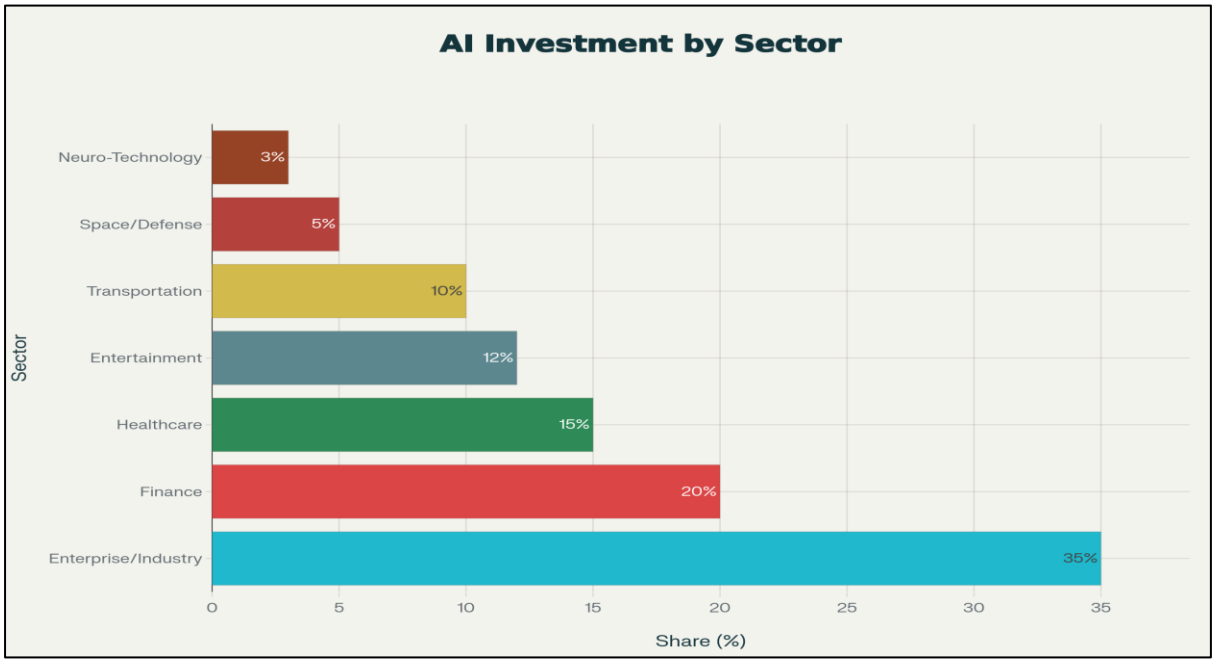
Alongside enthusiasm and massive investment, a competing narrative has emerged questioning whether AI hype exceeds realistic capabilities, the "**AI bubble**" thesis. Critics argue that generative AI systems, despite impressive performance on certain benchmarks, face fundamental limitations preventing the transformative applications enthusiast's project.

From the islands, I find myself in an unusual position regarding the AI bubble debate. On the mainland, people argue about whether AI hype is justified and whether certain predicted applications will materialize. Here, I think we're asking different questions entirely: We're not asking whether AI will transform the world. We're asking whether the technological transformation happening globally will include us, or whether it will deepen our marginalization.

The bubble narrative contains important truth: widespread exaggeration of near-term capabilities does characterize contemporary AI discourse. Yet the framing can obscure how AI hype generates productive effects, at least for some communities. Massive investment in AI infrastructure, even if some specific applications fail to materialize, creates real advances in computational capability, enables genuine experimental development, and funds substantial algorithmic progress. **But these advances occur primarily in wealthy nations and corporate contexts.**

The socio-technical insight is that **complementarity** between **financial exuberance** and **technical reality** need not be perfect to generate substantial progress, but only if you have access to the infrastructure and institutions where that progress is happening. Even if 80% of AI investment fails to produce promised returns, the 20% that succeeds could represent genuine breakthroughs. The more immediate question is: how do we ensure that AI development includes us, or at minimum, doesn't further marginalize us?





AI Bubble Perception and Sectoral Distribution of AI Investment

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5. CONTESTED TERRAIN OF ALGORITHMIC CONTROL

Kellogg, Valentine, and Christin provide crucial insights into how algorithms reshape workplace control and labour relations. Their framework identifies mechanisms through which employers use algorithms to control workers, reorganizing the fundamental terrain of control in workplaces. Algorithms can centralize decision-making by encoding managerial preferences into scoring systems; they can simultaneously enable new forms of worker resistance, as employees learn to game metrics or collectively challenge algorithmic rules.

In the context of the islands, I observe how government administrative algorithms replicate and entrench existing power asymmetries rather than addressing them. The administration has implemented algorithmic systems for welfare distribution, employment allocation, and resource management. On paper, these systems promise objectivity, efficiency, and reduced corruption. In practice, I've observed how they:

Assume universal documentation: Tribal populations who have relied on oral records rather than formal documentation find themselves excluded from algorithmic eligibility assessments.

Encode historical data biases: Algorithms trained on historical distributions of resources perpetuate patterns where certain communities (lower castes, tribal groups, women) received fewer resources previously, and the algorithm replicates these historical inequalities as though they represent inevitable patterns rather than the result of prior discrimination.

Depend on infrastructure that doesn't exist equally: Digital welfare distribution systems require bank accounts and mobile phones—infrastructure that remains unevenly distributed. Rural and tribal residents on remote islands may lack reliable phone connectivity, making participation impossible.

Create new forms of opacity: When welfare decisions are made by algorithmic systems, transparency about decision-making disappears. A tribal elder affected by an algorithmic denial of benefits cannot understand or challenge the decision in the way they could challenge a bureaucrat's discretionary judgment. The algorithm becomes a mechanism for hiding power rather than making it more transparent.

From my student life on the islands, I've experienced this algorithmic governance directly. The university's admission system uses algorithmic allocation based on quantitative criteria (board exam scores, entrance exam performance). While appearing merit-based, this system

marginalizes students from remote island schools whose educational infrastructure remains inferior. A student with equivalent capability but from a well-resourced mainland school appears "more meritorious" because quantitative criteria advantage those with better preparatory resources. The algorithm doesn't see educational inequality. It sees test scores and ranks accordingly, perpetuating advantage.

This is precisely the kind of complementarity failure visible when technological integration proceeds faster than social and infrastructural adaptation.

5.1. The Data Dimension: Who Is Represented in Training Data?

AI systems' dependence on data for training and operation embeds social structures directly into technical systems. Training data reflects historical distributions, images labelled with object categories encode the photographic practices, subject selection, and framing choices of thousands of photographers, predominantly from wealthy nations. Text data encodes linguistic patterns reflecting social hierarchies, power structures, and exclusions.

In the Islands, entire communities and their experiences are absent from the datasets that train global AI systems. The Shompen, Nicobarese, Onge, Andamanese, Jarawa, and other tribal populations have virtually no representation in internet datasets. Imagery of island ecosystems, tribal landscapes, or traditional practices is sparse. Language models trained predominantly on English, Mandarin, Spanish, and other major languages have minimal exposure to Andamanese, Nicobarese, or other island languages. This means AI systems trained on global data will be radically uninformed about island contexts, island governance challenges, or island community needs.

When models trained on such geographically biased data demonstrate bias (as documented extensively in facial recognition, loan approval, and hiring decision algorithms), the bias does not originate in algorithms but in the social structures reflected in training data. The underrepresentation of tribal faces in facial recognition training data causes these systems to perform poorly on tribal individuals. The lack of island-specific data in climate models causes these systems to give inadequate attention to island-specific vulnerability. The absence of island contexts in disaster response datasets means algorithms trained for disaster response perform poorly in island settings.

This section explains why AI is not pure technology by the means of looking at Socio-technical realities through multiple lens and my personal experiences.

6. AI AS A SOCIO-TECHNICAL SYSTEM

6.1. Technical Dimensions

The technical dimension of AI encompasses **mathematical frameworks** (linear algebra, probability theory, optimization algorithms); **computational architectures** (neural networks, attention mechanisms, transformer models); **specialized hardware** (GPUs, TPUs, neuromorphic chips); **software infrastructure** (machine learning frameworks like PyTorch and TensorFlow); and **data processing pipelines**.

This dimension is genuinely transformative. Breakthroughs in deep learning over the past decade, particularly the transformer architecture enabling large language models, represent real technical advances enabling novel capabilities. Yet these advances are concentrated in wealthy nations with computational infrastructure to support them. For peripheral regions, the technical dimension remains largely inaccessible**.**

6.2. Social Dimensions: Geography, Institutions, and Power

The social dimension encompasses: **organizational structures** coordinating work between humans and machines; **labour arrangements** allocating tasks and compensation; **institutional frameworks** (regulatory agencies, professional associations, educational institutions) establishing standards and legitimacy; **financial mechanisms** allocating capital toward specific technical directions; **geographic distributions** of infrastructure and opportunity; and **cultural meanings** attributing significance to AI and its potential impacts.

The social dimension proves equally important to the technical dimension, and it is precisely where geographic marginalization becomes visible. No algorithm becomes deployed without organizational decisions about which problems to address, and those decisions are made by people in wealthy centers serving wealthy markets. No training data gets collected without institutional practices of measurement and extraction, and those practices preferentially capture data from regions with digital infrastructure. No AI system proves effective without social acceptance and integration into existing institutional practices, yet integration is assumed rather than negotiated with communities.

6.3. Joint Optimization: The Complementarity

From a socio-technical perspective, AI's trajectory depends on achieving "joint optimization", designing technical systems and social contexts to work together **productively,** with particular attention to geographic and infrastructural realities. This requires:

Organizational Alignment: Firms and governments deploying AI must simultaneously restructure workflows, redefine roles, retrain workers, and modify performance metrics in ways attentive to local organizational capacity. Technical excellence proves insufficient without organizational change, yet organizational change must be co-designed with the communities it affects.

Governance Frameworks: Effective AI governance requires complementarity between technical standards (how algorithms are validated, audited, and monitored) and social institutions (regulatory agencies, oversight bodies, liability frameworks) enforcing accountability, with particular attention to how governance frameworks can account for geographic and infrastructural variation.

Data Justice: Managing data as socio-technical problem requires mechanisms ensuring fair data collection, transparent data use, benefit-sharing with data subjects, and redress for harms caused by data-driven systems, particularly ensuring that data extraction from peripheral communities translates into benefits for those communities, not merely benefits for distant corporations.

Infrastructure Development: Joint optimization requires that technical and organizational systems are deployed only when underlying infrastructure—electricity, connectivity, institutional capacity, can support them. Yet I observe the opposite pattern: technology is deployed before infrastructure is adequate, then blamed for "lack of adoption" when predictably it underperforms.

Skill Development: AI's productive potential depends on complementary development of human expertise, data science talent, domain specialists understanding specific applications, ethicists identifying problems, and workers adapting to transformed task environments, trained and supported within peripheral communities, not just imported from centers.

Genuine Participation: Effective AI governance requires community comprehension of what AI systems can and cannot do, enabling informed participation in decisions about their deployment. **Yet on the islands, major technological integrations proceed without community consultation, education about what systems do, or space for community input.**

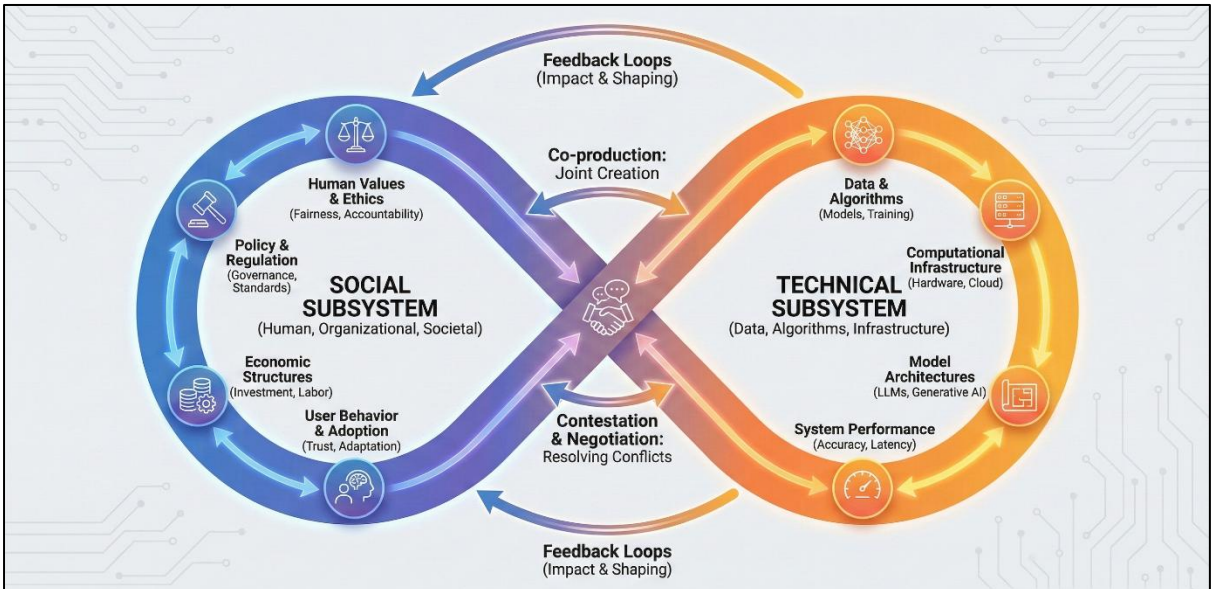

Figure 4: The Socio-Technical System of AI. This infographic visualizes AI as a complex system of interacting social and technical components, illustrating how they co-produce and influence each other through continuous feedback and negotiation.

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7. CONCLUSION: TOWARDS GEOGRAPHICALLY ATTENTIVE AND EQUITABLE AI DEVELOPMENT

The examination of AI as socio-technical system, viewed through my lived experiences and observations of peripheral island communities, yields several crucial implications:

First, AI cannot be reduced to either technical or social components, understanding AI requires analyzing their necessary interaction, **with particular attention to how this interaction varies across geographies**. Technical expertise alone proves insufficient for responsible AI development; equally, social governance divorced from technical understanding risks ineffectiveness, and governance designed from wealthy-center perspectives inevitably marginalizes peripheral realities.

Second, the trajectory of AI development is not technically predetermined but emerges from specific institutional, financial, and organizational choices that systematically privilege wealthy geographies. The concentration of AI capability among large technology firms in wealthy nations reflects not technical necessities but financial structures enabling massive R&D investment, data accumulation, and computing infrastructure control among dominant platforms in those locations. Alternative trajectories were possible, distributed development, locally rooted AI, technology designed from peripheral perspectives, but specific institutional arrangements foreclosed these options.

Third, complementarity between technical and social dimensions is not automatic but requires deliberate design and institutional innovation, **including institutional innovation specifically attentive to geographic and infrastructural realities**. Massive investments in AI hardware and algorithmic research prove insufficient for transformative social benefits unless complemented by organizational restructuring, workforce development, equitable governance frameworks, infrastructural development in peripheral regions, and institutional arrangements ensuring broad benefit distribution.

Fourth, the "AI bubble" debate itself requires socio-technical analysis from peripheral perspectives. From the islands, the question is not whether hype is justified globally, but whether peripheral communities will be included in whatever benefits AI ultimately generates, or whether we will be further marginalized as technology concentrates in wealthy centers.

Finally, understanding AI as socio-technical system suggests governance approaches emphasizing joint optimization rather than either purely technical solutions (algorithmic fairness, interpretability) or purely regulatory approaches (restricting AI development). Responsible AI development requires complementary advances in technical sophistication, institutional sophistication, social acceptance, ethical clarity, and crucially, geographic equity and infrastructural development.

The future of AI is not primarily a technical question but a political economy question about who gets to participate in defining and developing AI, whose knowledge gets incorporated, whose needs drive technical directions, and how the benefits and harms get distributed across geographies.

The technical capability exists for transformative applications. The financial resources exist for massive investment. What remains uncertain is whether institutions can evolve rapidly enough to ensure these capabilities serve broad social benefit, particularly benefit for the world's peripheral, marginalized, and geographically isolated communities, rather than merely concentrating power, knowledge, and opportunity among dominant firms and wealthy nations.

The ferry continues to connect these islands to the mainland, carrying people, goods, and technological dreams. Yet without addressing the fundamental infrastructural, institutional, and epistemic gaps that distinguish island from mainland, the AI revolution will arrive by ferry to isolated communities, unmoored from their needs, unattentive to their knowledge, and yet another iteration of technological integration that serves distant interests while leaving peripheral communities to adapt or be left behind.

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